

A Physics-Informed Hybrid Approach for Cyberattack Detection in the Power Grid

Fatemeh Sharifi, *Graduate Student Member, IEEE*, and Ali Mehrizi-Sani, *Senior Member, IEEE*

Abstract—Cyberattacks targeting critical infrastructure such as the power grid pose significant risks to the reliability and security of energy networks. Among the cyberattacks, false data injection (FDI) attacks, which manipulate power grid data, is a serious concern due to their potential to cause widespread disruptions and damage. The 2015 Ukraine power grid cyber-attack demonstrated its destructive potential and revealed the real-world impact of such threats. It is especially challenging to detect AC FDI attacks since they are based on the AC power flow. Traditional cyberattack detection methods are primarily based on rule-based or statistical anomaly detection. They often fall short in detecting these cyber threats due to their design based on static thresholds. Recent advances in deep learning show promise in improving detection accuracy and adaptability. Building on these advances, this paper proposes a novel physics-informed approach for detecting AC FDI attacks in the power grid, integrating graph attention convolutional networks (GACN), long short-term memory (LSTM) networks, and unscented Kalman filters (UKF). GACN uses the physical topology of the power grid to capture spatial dependencies and correlations between grid components, while LSTM models temporal dynamics. In addition, UKF enhances the detection capabilities of the proposed method as a second layer of protection. Extensive simulation studies using PSCAD/EMTDC datasets demonstrate that the proposed method outperforms several baseline models in detecting sophisticated cyber threats in the power grid. These baseline models encompass a range of simpler machine learning and deep learning architectures.

Index Terms—AC false data injection, cyberattack, graph attention convolutional neural network, long short-term memory, physics-informed detection, unscented Kalman filter.

I. INTRODUCTION

In recent years, cyberattacks targeting critical infrastructure systems, such as the power grid, pose significant threats to the reliability and security of energy networks [1]–[3]. Among the spectrum of cyber threats, false data injection (FDI) attacks represent a critical vulnerability, wherein adversaries alter power grid data. FDI attacks can lead to erroneous decisions and potentially catastrophic consequences [4], [5]. A particular variant, AC FDI attacks, is especially challenging to detect due to its design based on the AC power flow [6]–[8]. Detecting and mitigating AC FDI attacks in the power grid is important to ensure its integrity and resilience.

Traditional approaches to FDI attack detection often rely on rule-based methods or statistical anomaly detection techniques, which may struggle to adapt to the evolving nature of cyber

threats and the complex dynamics of the power grid [12], [13]. To address these challenges, recent research has explored the application of advanced machine learning and deep learning techniques for FDI attack detection, aiming to enhance the accuracy and efficiency of detection algorithms [14]–[16]. The use of deep neural networks (DNN) is increasing due to their enhanced capacity for extracting information [17], [18].

In recent studies, various approaches are proposed to tackle FDI detection by DNN. [19] addresses the growing risk of cyber intrusions in power grid digitalization using an intrusion detection and mitigation system (IDMS) employing DNNs. The proposed scheme in [20] employs a conditional deep belief network (CDBN) to effectively identify the high-dimensional temporal behavior features of FDI attacks. An FDI attack detection is achieved through the integration of discrete wavelet transform (DWT) and DNN techniques to construct an intelligent detection system in [21]. In [22], the authors propose a deep learning-based remedial action scheme (RAS) utilizing long short-term memory (LSTM) networks to mitigate the effects of FDI cyberattacks on the power grid. A deep reinforcement learning-based method is proposed in [23] to identify vulnerabilities in load frequency control of the electric grid, synthesizing FDI and load switching attacks. In [11], a two-step approach is proposed, combining convolutional neural network with long short-term memory (CNN-LSTM) models and an unscented Kalman filter (UKF) to detect DC FDI attacks on the IEEE 39-bus system.

However, as highlighted in [1], [24], [25], the utilization of graph neural networks (GNN) holds great promise for modeling complex relationships and dependencies in networked systems, making them well-suited for physics-informed FDI attack detection. In [26], the authors propose an unsupervised approach combining dual graph-convolutional autoencoder (DAE) and generative adversarial network (GAN) techniques. The fault diagnosis method in [27] integrates graph convolutional networks (GCN) with prior system knowledge and measurement data, offering improved diagnosis performance by combining structural analysis and GCN techniques. The approach in [9] employs short-time Fourier transform (STFT) and a two-channel convolutional neural network (2C-CNN) for spectral-temporal correlation modeling, followed by GCN for spectral-spatial relationship modeling using nodal admittance matrix and physical properties of the power grid to localize FDI attacks. However, this method lacks the ability to fully exploit the temporal dynamics of the system and may struggle with real-time adaptability.

Moreover, current deep learning-based methods for FDI attacks encounter challenges in interpretability, often characterized as “black box” approaches because of the inherent

F. Sharifi and A. Mehrizi-Sani are with the Bradley Department of Electrical and Computer Engineering, Virginia Tech, Blacksburg, VA, 24061, USA (e-mails: {fatima94,mehrizi}@vt.edu).

This work is supported in part by the National Science Foundation (NSF) under award ECCS-1953213 and in part by the State of Virginia's Commonwealth Cyber Initiative (www.cyberinitiative.org).

TABLE I
RECENT STUDIES ON FDI DETECTION BY DNNs

Paper	Detection Method	Physics-Based Graph	Spatial Pattern Recognition	Temporal Pattern Recognition	Attention Mechanism	Second Layer of Protection	AC FDI
[9]	2C-CNN + GCN	✓	✓	✓	✗	✗	✓
[10]	DAMGAT	✓	✓	✗	✓	✗	✗
[11]	CNN-LSTM + UKF	✗	✓	✓	✗	✓	✗
This paper	GACN-LSTM + UKF	✓	✓	✓	✓	✓	✓

complexity of the models [10]. Therefore, using the attention mechanism which attains model interpretability by the quantification of attention weights throughout the learning process can address this concern. By incorporating a dual-attention mechanism into a multi-head graph attention network (DAMGAT), [10] captures spatial correlations between attack characteristics and measurement data. However, using GCN instead of GNN has benefits for processing graph-structured data, efficient propagation of information through the graph, and scalability for handling large-scale power grid networks.

Recent work on the power grid increasingly emphasizes physics-informed learning, where data-driven methods are constrained by the underlying physical laws and network structure. For example, [28] introduces a physics-aware neural network for distribution system state estimation, [29] develops a physics-informed graphical learning approach for three-phase line parameter estimation, and [30] provides a comprehensive review of physics-informed neural networks for power system applications.

This paper proposes a novel physics-informed approach that explicitly incorporates the topology of the power grid. This method leverages the synergistic combination of graph attention convolutional networks (GACN), long short-term memory (LSTM) networks, and UKF. Our approach integrates physics-informed modeling at two levels: first, by embedding the grid topology and electrical variables into GACN, and second, by enforcing consistency with the AC measurement model through UKF. Therefore, by utilizing a graph-based model grounded in the system structure, the proposed framework captures both spatial and temporal correlations within the grid. As a second layer of detection, UKF has the advantage of effectively estimating the true state of a system based on noisy measurements and its dynamics. Moreover, UKF can adapt to changes in system dynamics over time, making it suitable for detecting attacks with less than 20% falsification.

The main contributions of this paper are

- Applying GACN for handling large-scale data and providing interpretability of the measurements in the power grid;
- Using LSTM networks to capture and analyze temporal patterns in the power grid;
- Implementing UKF as a complementary layer of security for improved detection capabilities; and
- Detecting and localizing multiple AC FDI attacks on random number of buses or specific area of the power grid.

The remainder of this paper is an overview of the proposed framework. Section II explains the architecture and methodology. Section III describes the training dataset and AC FDI attack preparation. Simulation details and results are presented in Section IV. Finally, Section V summarizes the findings and their significance.

II. METHODOLOGY

This section presents the methodology developed for detecting FDI attacks in the power grid. The proposed framework integrates three core components: GACN, LSTM networks, and UKF. The following subsections detail the function and integration of each component within the proposed framework.

A. Graph Attention Convolutional Network

GACN is employed to model the complex spatial dependencies and relationships in the power grid. It uses graph-based representations to capture the correlations between different nodes. This physics-informed model employs an attention mechanism to focus on the most relevant parts of the graph, improving both detection accuracy and model interpretability.

Let $G = (V, E)$ represent the power grid, where V denotes the set of nodes (buses), and E represents the edges (transmission lines). The feature matrix $X \in \mathbb{R}^{N \times F}$ describes the node features, where N is the number of buses, and F is the number of features per bus. In other words, this matrix encapsulates features such as voltage, phase angle, real power, reactive power, and other relevant properties linked to each bus within the graph.

GACN computes the node embeddings $H \in \mathbb{R}^{N \times F}$ by aggregating information from neighboring buses using an attention mechanism. Here, the embedding matrix H represents the updated bus features and captures the transformed representations of each bus after aggregation.

$$H_i = \sigma \left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij} W X_j \right), \quad (1)$$

where H_i represents the embedding of bus i , $\mathcal{N}(i)$ denotes the neighborhood of bus i , W is the learnable weight matrix, σ is a nonlinear activation function (ReLU), and α_{ij} is the attention coefficient between bus i and j which determines the importance of neighboring buses. The attention coefficients α_{ij} are computed as

$$\alpha_{ij} = \frac{\exp(\text{LeakyReLU}(a^T [W X_i \| W X_j]))}{\sum_{k \in \mathcal{N}(i)} \exp(\text{LeakyReLU}(a^T [W X_i \| W X_k]))}, \quad (2)$$

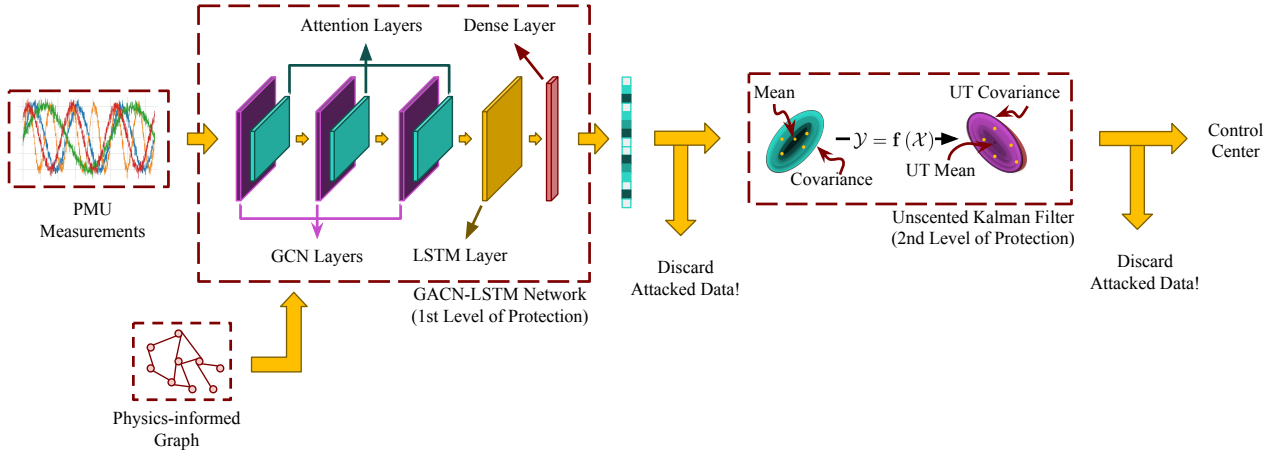


Fig. 1. Schematic diagram of the proposed method, illustrating the two layers of protection and the detailed components within each layer.

where a is a learnable vector, and $\parallel$ denotes concatenation. The attention mechanism allows the model to dynamically focus on the most informative connections in the graph.

B. Long Short-Term Memory Networks

To capture the temporal dependencies in power grid data, LSTM is integrated into the framework. LSTM is well-suited for sequential data analysis and can effectively model long-term dependencies, which are critical for detecting temporal patterns indicative of FDI attacks. Given a sequence of node embeddings $\{H_t\}_{t=1}^T$ obtained from GACN over time T , LSTM processes this sequence to learn temporal patterns. LSTM maintains a hidden state h_t and a cell state c_t at each time step t , updated according to

$$\begin{aligned} i_t &= \sigma(W_i H_t + U_i h_{t-1} + b_i), \\ f_t &= \sigma(W_f H_t + U_f h_{t-1} + b_f), \\ o_t &= \sigma(W_o H_t + U_o h_{t-1} + b_o), \\ c_t &= f_t \odot c_{t-1} + i_t \odot \tanh(W_c H_t + U_c h_{t-1} + b_c), \\ h_t &= o_t \odot \tanh(c_t), \end{aligned} \quad (3)$$

where i_t, f_t, o_t are the input, forget, and output gates, respectively; W, U , and b are learnable parameters; σ denotes the sigmoid activation function; and $\odot$ represents element-wise multiplication. By processing the sequence of node embeddings, LSTM captures the temporal dynamics and correlations in power grid data, enhancing the detection of temporal patterns associated with FDI attacks.

C. Unscented Kalman Filter

UKF extends the conventional Kalman filter to effectively handle nonlinear systems. Unlike the extended Kalman filter (EKF), which relies on linear approximations via first-order Taylor expansions, UKF maintains a more accurate representation of the probability distribution utilizing a specific set of ‘‘sigma points’’ to characterize the distribution of the system states. These sigma points are passed through the nonlinear system dynamics, enabling UKF to preserve the statistical properties of the original distribution more accurately compared to the linearization technique of the EKF. By calculating

the mean and covariance from the propagated sigma points, UKF captures nonlinearities in the system more effectively. This deterministic sampling approach makes UKF particularly advantageous for systems with significant nonlinear behaviors or where precise linearization is difficult [31]. In addition, unlike deep Kalman filter (DKF), which requires learning complex latent dynamics and incurs substantial computational and training costs, UKF directly incorporates the measurement models of the power grid. Due to this feature of UKF, our hybrid method is able to combine the advantages of both data-driven and measurement-based approaches. For further exploration of transient power grid state estimation using UKF, see [32], [33].

To identify anomalies in the data, the chi-squared (χ^2) test is employed [34]. χ^2 is calculated by summing the squared differences between observed and expected values, normalized by the expected measurement. A lower χ^2 value indicates a less substantial gap between observed and expected measurements, signaling a stronger correlation between variables or a reduced likelihood of abnormal data.

A threshold value γ is defined to determine the probability that a measurement lies within the uncompromised region. This probability-based threshold is calculated as

$$V_\gamma(k) = \{y_k : (y_k - \hat{y}_k)^T S(k)^{-1} (y_k - \hat{y}_k) \leq \gamma\}, \quad (4)$$

where y_k denotes the measurement, and $S(k)$ represents the innovation covariance matrix at time step k .

The optimal selection of the threshold and innovation covariance matrix requires careful tuning to balance sensitivity and false alarm rate. Achieving this balance is important for designing an effective algorithm to detect abnormal data patterns. The anomaly detection threshold γ is optimized through a two-step process. First, an initial value is set based on the inverse χ^2 distribution at a 95% confidence level, with the degrees of freedom equal to the dimension of the innovation vector. This statistical choice corresponds to a nominal false-alarm rate under the assumed Gaussian noise model. Second, a grid search is performed on the validation dataset, sweeping γ across a range from the 95% to the 99.9% quantiles. For each candidate value, the detection performance

is evaluated in terms of the F1-score and recall, while ensuring the false-positive rate remains within operational limits. The value of γ that provides the best trade-off between high detection accuracy and low false-alarm rate is selected for the final model. This procedure ensures that the UKF's statistical decision boundary is both theoretically grounded and tuned for practical effectiveness. The UKF algorithm with anomaly detector is explained in Algorithm 1.

“Physics-informed” means that the model structure and its computations are constrained by the power grid physics: (i) the graph and neighborhood system used by GACN are built from the bus-branch topology, so message passing occurs only along physically connected elements; (ii) node features contain electrical variables (voltage magnitude/angle, line real and reactive power), so attention is learned over physically meaningful quantities; and (iii) the second layer (UKF) embeds the AC measurement model and uses the chi-squared residual test consistent with state estimation practice. Together, these design choices carry the power grid's physical constraints rather than purely data-driven correlations.

Algorithm 1 UKF Algorithm with Anomaly Detector

Initialize: Initial state $\mathbf{m}_0 = m$, covariance $\mathbf{P}_0 = P$, tuning parameters α, β, κ , anomaly threshold γ

Output: Updated state $\mathbf{m}_k$, covariance $\mathbf{P}_k$, anomaly detection status

Compute $\lambda = \alpha^2(n + \kappa) - n$

for $k = 1, 2, \dots, T$ **do**

Prediction:

 Generate sigma points $\mathcal{X}_{k-1}^{(i)}$ from $\mathbf{m}_{k-1}$ and $\mathbf{P}_{k-1}$

 Propagate sigma points through the dynamics:

$$\hat{\mathcal{X}}_k^{(i)} = \mathbf{f}(\mathcal{X}_{k-1}^{(i)})$$

 Compute predicted state $\mathbf{m}_k^-$ and covariance $\mathbf{P}_k^-$

Update:

 Transform sigma points for measurement:

$$\hat{\mathcal{Y}}_k^{(i)} = \mathbf{h}(\hat{\mathcal{X}}_k^{(i)})$$

 Compute predicted measurement $\boldsymbol{\mu}_k$ and innovation covariance $\mathbf{S}_k$

 Calculate cross-covariance $\mathbf{C}_k$

 Compute Kalman gain: $\mathbf{K}_k = \mathbf{C}_k \mathbf{S}_k^{-1}$

 Update state estimate: $\mathbf{m}_k = \mathbf{m}_k^- + \mathbf{K}_k(\mathbf{y}_k - \boldsymbol{\mu}_k)$

 Update covariance estimate: $\mathbf{P}_k = \mathbf{P}_k^- - \mathbf{K}_k \mathbf{S}_k \mathbf{K}_k^\top$

Anomaly Detection:

 Calculate the residual $\mathbf{y}_k = \mathbf{y}_k - \boldsymbol{\mu}_k$

 Compute the chi-squared test statistic: $\chi^2 = \mathbf{y}_k^\top \mathbf{S}_k^{-1} \mathbf{y}_k$

if $\chi^2 > \gamma$ **then**

Anomaly detected at time step k

else

 No anomaly detected

end if

end for

III. DATA PREPARATION IN FALSE DATA INJECTION

A. Dataset Collection Using PSCAD/EMTDC Software

To evaluate the effectiveness of the proposed approach, the IEEE 118-bus and 39-bus systems are used as the testbed.

These systems are chosen due to their complexity and ability to model real-world power grid scenarios.

The dataset is constructed by simulating a variety of operational conditions using PSCAD/EMTDC soft tool. The dataset includes measurements of bus voltage magnitudes, bus phase angles, line real power, and line reactive power. Data collection is automated using a Python script that runs PSCAD/EMTDC simulations across a range of predefined scenarios. In each scenario, a disturbance i for $i \in \{\text{fault, load change, generator disconnection}\}$ is introduced at bus j for $j \in \{1, 2, \dots, N\}$ at time step k for $k \in \{1, 2, \dots, T\}$. N denotes the total number of buses, and T represents the total number of simulation time steps. Fault scenarios include three types: ABC-to-ground, A-to-B, and A-to-ground. By randomly varying the disturbance type, location, and occurrence time, a comprehensive dataset is generated. This dataset captures a broad range of potential system states and conditions to ensure generalizability in model evaluation.

To determine the number of samples needed for effective model training, a learning-curve analysis is performed by incrementally increasing the size of the training set and monitoring the F1-score and recall on a fixed validation set. The performance improves significantly as the training set increases from 20% to about 80% of the full dataset, after which the gains become marginal (less than 1% change in F1-score). Based on this observation, the full dataset—consisting of 1,200 samples for the IEEE 118-bus system and a proportionally similar number for the IEEE 39-bus system is considered. This dataset ensures the model has adequate exposure to diverse operational scenarios, disturbance types, and attack patterns to generalize effectively.

The final dataset for the IEEE 118-bus system consists of 1,200 samples, each containing measurements from 118 buses across 4 different parameters. Each sample is recorded over 1,500 time steps, capturing the dynamic behavior of the power grid during each scenario with a time resolution of 0.001 seconds per step. The phase angles are measured in radians, while the voltage magnitudes, real power, and reactive power are in per unit (pu). The base apparent power is 100 MVA, and the base voltage levels are 138 or 345 kV, depending on the area of the grid being analyzed. The same procedure is done to collect data for the IEEE 39-bus system.

To ensure effective training of deep learning models, min-max normalization is applied to standardize the raw data. This normalization process scales all features to a uniform range, typically $[0, 1]$, enhancing the convergence of learning algorithms and improving model generalization across different scenarios. In other words, this preprocessing step is needed to mitigate the effects of varying data scales and ensure that the model learns effectively from the input data.

B. False Data Injection Attack with AC Power Flow

FDI attacks pose a significant threat to power grid security by intentionally altering sensor measurements to mislead control systems, potentially leading to erroneous operational decisions and disruptions. These attacks exploit vulnerabilities in measurement data, making it challenging to distinguish

between legitimate and manipulated data. For instance, an FDI attack could involve altering voltage measurements of certain buses to falsely indicate a stable grid condition, causing the control system to delay necessary corrective actions and potentially leading to grid instability or outages.

The AC power flow is governed by nonlinear equations representing the steady-state operation of the power grid. Therefore, AC FDI attacks are particularly challenging to detect by bad data detection algorithms due to their alignment with AC power flow. The AC power flow equations show the relationship between bus voltages and the real and reactive power flows in the network. For a given bus i , the real power P_i and reactive power Q_i are defined by

$$P_i = \sum_{j=1}^n V_i V_j (G_{ij} \cos \theta_{ij} + B_{ij} \sin \theta_{ij}), \quad (5)$$

$$Q_i = \sum_{j=1}^n V_i V_j (G_{ij} \sin \theta_{ij} - B_{ij} \cos \theta_{ij}), \quad (6)$$

where P_i and Q_i represent the real and reactive power injections at bus i , respectively. V_i is the voltage magnitude at bus i . $\theta_{ij} = \theta_i - \theta_j$ is the voltage angle difference between buses i and j . G_{ij} and B_{ij} are the conductance and susceptance between bus i and bus j , respectively.

The nonlinearity in these equations stems from the trigonometric terms, which complicates accurate state estimation. In an AC FDI attack, an attacker manipulates the sensor measurements that feed into these equations, introducing errors that lead to incorrect state estimations and compromised system control.

AC FDI attacks can be modeled as additive perturbations to the original measurements. If z represents the vector of true measurements, an AC FDI attack modifies these measurements to produce a compromised measurement vector z_a

$$z_a = z + a, \quad (7)$$

where z_a denotes the compromised measurement vector, z is the true measurement vector, and a is the attack vector that represents the malicious modifications introduced by the attacker. The weighted least squares method is used to calculate the system state estimate:

$$x = \min_x [z - h(x)]^T R^{-1} [z - h(x)], \quad (8)$$

where the system state is denoted by x , while the estimated system state that best aligns with the measurement data z is denoted by $\hat{x}$. The nonlinear relationship between measurements and states is described by function h .

To conduct AC FDI attacks, a must be selected so that the residual r_a is below a threshold τ , usually set by an operator.

$$r_a = \|z_a - h(x_a)\|_2^2. \quad (9)$$

Therefore, a successful FDI attack can manipulate power grid measurements in a way that the AC state estimator fails to identify the bad data, effectively disguising the attack as accurate measurements. In other words, an assumption is made that the attacker knows the network model and AC measurement function $h(\cdot)$ sufficiently to craft an attack vector

a whose residual r_a stays below the operator's threshold τ . The goal of the proposed methodology is to detect the presence of the attack vector a to maintain the integrity and security of the power grid. Effective detection requires distinguishing between natural measurement variations and those caused by malicious activities.

IV. PERFORMANCE EVALUATION

This section presents the performance evaluation of the proposed approach for detecting AC FDI attacks in the power grid. The main objectives of this evaluation are to evaluate detection accuracy, computational efficiency, and robustness under various attack scenarios. Additionally, the proposed method is compared against several baseline models to demonstrate its superiority in terms of different performance metrics.

A. Attacks Generation

AC FDI attacks are randomly conducted on the buses, where the number of affected buses is uniformly distributed within the interval $[1, N/6]$, with N is the total number of buses. The process for generating AC FDI attacks is described in Section III. Since AC FDI attack detection is treated as a multi-label classification problem, data scaling attacks are added to the dataset to create a more evenly distributed attack dataset. For data scaling attacks, both the number of targeted buses and the scale factor are uniformly distributed. The number of targeted buses falls within the range $[1, N/2]$, while the scale factor is chosen from the interval $[0.9, 0.95] \cup [1.05, 1.1]$. Attack types are gathered in a compact table with parameter ranges.

In practice, AC FDI attacks do not always occur uniformly across all buses, which can lead to an imbalanced dataset where some buses have very few attack samples. To address this, data-scaling attacks are introduced as an additional class of disturbances. In these attacks, both the number of targeted buses and the scaling factor are randomly varied, which increases the diversity of attack patterns. Therefore, the dataset becomes more balanced across different buses and scenarios, ensuring that the model does not become biased toward buses with more frequent attack samples. This balancing step improves the robustness by giving the algorithm adequate exposure to a wider range of possible attack conditions. The complete distribution of disturbances and attack scenarios used in this study is summarized in Table IV.

B. Evaluation Metrics

The evaluation of the proposed model is conducted using several key performance metrics, which are elaborated in detail below [10].

- **Accuracy:** The ratio of correctly identified instances (both positive and negative) to the total number of instances. This metric provides a general measure of the model's correctness.
- **Precision:** The proportion of correctly identified positive instances to the total instances predicted as positive.

TABLE II
PERFORMANCE COMPARISON OF THE PROPOSED MODEL AND BASELINE MODELS IN THE FIRST ATTACK SCENARIO

Model	IEEE 39-Bus System				IEEE 118-Bus System			
	Accuracy	Precision	Recall	F1-Score	Accuracy	Precision	Recall	F1-Score
SVM	0.610	0.595	0.496	0.526	0.584	0.471	0.356	0.395
CNN	0.694	0.714	0.581	0.602	0.642	0.614	0.494	0.526
CNN-LSTM [11]	0.716	0.735	0.680	0.664	0.684	0.721	0.677	0.619
CNN-LSTM + UKF [11]	0.786	0.865	0.810	0.844	0.764	0.821	0.777	0.819
GACN	0.864	0.902	0.878	0.887	0.849	0.899	0.849	0.834
GACN-LSTM	0.876	0.925	0.915	0.904	0.863	0.918	0.910	0.898
Proposed Model	0.912	0.931	0.920	0.918	0.909	0.924	0.916	0.904

TABLE III
PERFORMANCE COMPARISON OF THE PROPOSED MODEL AND BASELINE MODELS IN THE SECOND ATTACK SCENARIO

Model	IEEE 39-Bus System				IEEE 118-Bus System			
	Accuracy	Precision	Recall	F1-Score	Accuracy	Precision	Recall	F1-Score
SVM	0.608	0.590	0.491	0.523	0.580	0.467	0.348	0.393
CNN	0.687	0.706	0.575	0.593	0.634	0.613	0.490	0.517
CNN-LSTM [11]	0.712	0.727	0.671	0.664	0.680	0.720	0.670	0.610
CNN-LSTM + UKF [11]	0.778	0.861	0.806	0.843	0.759	0.815	0.773	0.818
GACN	0.864	0.901	0.875	0.884	0.847	0.896	0.839	0.831
GACN-LSTM	0.872	0.924	0.912	0.895	0.853	0.917	0.903	0.895
Proposed Model	0.906	0.924	0.910	0.915	0.902	0.915	0.910	0.898

TABLE IV
SAMPLE DISTRIBUTION ACROSS DISTURBANCE TYPES AND ATTACK SCENARIOS

Scenario Category	Scenario Type	Parameter Range	IEEE 39-Bus Count	IEEE 118-Bus Count
Disturbances	Three-phase fault (ABC-G)	Random bus, random time	120	360
	Line-to-line fault (A-B)	Random bus, random time	120	360
	Single-phase fault (A-G)	Random bus, random time	120	360
	Load change	$\pm 10\%$ to $\pm 15\%$ change in load	120	360
	Generator disconnection	Random generator outage	120	360
AC FDI Attacks	Load-bus attack	Mag.: $[-0.15, -0.1] \cup [0.1, 0.15]$ pu; buses: $[1, N/6]$	120	360
	Generator-bus attack	Mag.: $[-0.15, -0.1] \cup [0.1, 0.15]$ pu; buses: $[1, N/6]$	120	360
	Combined attack	Mag.: $[-0.15, -0.1] \cup [0.1, 0.15]$ pu; buses: $[1, N/6]$	120	360
	Data scaling attack	Scale: $[0.9, 0.95] \cup [1.05, 1.1]$; buses: $[1, N/2]$	120	360

Precision is crucial for understanding the model's ability to correctly identify actual AC FDI attacks without generating excessive false positives.

- Recall (Sensitivity): The proportion of correctly identified positive instances to the total actual positive instances. This metric reflects the model's ability to detect attacks and is vital for evaluating its effectiveness in preventing undetected attacks.
- F1-Score: The harmonic mean of precision and recall, providing a single measure that balances both concerns. This metric is especially useful when dealing with class imbalance or when both false positives and false negatives need to be minimized.

These metrics provide a comprehensive view of the model's performance, allowing the evaluation to assess not only the model's accuracy but also its reliability, efficiency, and suitability for real-world applications.

C. Baseline Comparisons

To validate the effectiveness of the proposed approach, comparisons are made against several baseline models for AC FDI attack detection in the power grid. These baseline models are selected to provide a comprehensive comparison across different machine learning paradigms. These baseline models include: support vector machine (SVM), convolutional neural network (CNN), convolutional neural network with long short term memory (CNN-LSTM), CNN-LSTM with UKF proposed in [11], graph attention convolutional network (GACN), and GACN-LSTM.

To ensure a fair and informative comparison, the baseline models are selected to represent a range of complexity and capability in handling spatial, temporal, and physics-based dependencies in the power grid. Simpler methods such as support vector machines (SVM) and convolutional neural networks (CNN) capture static or spatial features but lack explicit temporal modeling. The CNN-LSTM baseline extends this by incorporating temporal dependencies, while CNN-LSTM+UKF in [11] further integrates an estimator.

Graph attention convolutional networks (GACN) capture the grid's physical topology, and GACN-LSTM combines spatial and temporal modeling. Including these baselines allows a stepwise comparison, showing the effect of adding temporal modeling, graph-based spatial modeling, and estimation. Other classical machine learning algorithms such as k-nearest neighbors (KNN) and naive Bayes (NB) are excluded from the reported results because they cannot effectively represent graph-structured or temporal dependencies, and initial screening experiments showed substantially lower performance than the selected baselines.

D. Setup and Model Architecture

All simulation studies are conducted using Python 3.8.10, along with the Pypower 5.1.17, Tensorflow 2.13.0, and Sklearn 1.3.2 libraries, on a system equipped with an Intel i7-10700 CPU and 16GB of RAM. The dataset is split into training, validation, and testing sets with a ratio of 6:2:2. Model hyperparameters are tuned using a random search strategy, and the model that achieves the highest recall on the validation set is selected.

To assess real-time feasibility, the inference latency and computational requirements of the proposed method are measured. The forward pass of the GACN-LSTM is profiled on the experimental setup with batch size one to emulate streaming operation. On the IEEE 118-bus dataset, the neural network forward pass requires approximately 2.1 ms per time step, dominated by graph attention convolutional layers and LSTM recurrence. The UKF update, which involves state prediction, covariance update, and residual calculation, adds 0.7 ms per time step, with complexity scaling as $O(n_x^3)$ where n_x is the state dimension (for IEEE-118, $n_x = 472$). The overall end-to-end latency is therefore around 2.8 ms per sample, which is well below the 16.7 ms budget required for real-time monitoring at a 60 Hz PMU reporting rate.

Memory requirements also remain modest. The trained GACN-LSTM model has about 1.3M parameters (~5.2 MB), and the UKF state and covariance matrices require less than 2 MB for the IEEE-118 system. These measurements indicate that the hybrid model can be deployed on commodity multi-core CPUs in control centers without dedicated GPUs, while still meeting real-time requirements for wide-area monitoring.

For the CNN-LSTM model, the architecture consists of three convolutional layers, each with a kernel size of 3×3 . A single LSTM layer follows the convolutional layers, before a dense layer whose dimension corresponds to the number of buses in the test case. For the GACN-LSTM model, the structure includes three graph convolutional layers with the same 3×3 kernel size, but each graph convolutional layer is followed by an attention mechanism. Similarly, a single LSTM layer is used before a dense layer, where the dimension is the number of buses in the test case.

Different combinations of these models were systematically evaluated to find the most effective approach for detecting AC FDI attacks. The optimal architecture was selected based on its ability to balance detection accuracy, computational efficiency, and resilience to different attack patterns. The chosen design

uses a layered approach where GACN first processes location-based grid features, allowing LSTM to then analyze how these features change over time. By analyzing both location-based and time-based patterns, this combination performs better than other model arrangements in tracking power grid state changes.

Both models are trained using the Adam optimizer with an initial learning rate of 0.001 and a batch size of 64. The training process continues for up to 100 epochs, with early stopping implemented if the validation performance shows no improvement for 5 consecutive epochs. The learning rate is dynamically adjusted throughout training. Both models use binary cross-entropy as their loss function.

E. Results and Analysis

All models are evaluated using online measurements obtained from the test cases simulated in PSCAD/EMTDC, with the testing and analysis carried out through Python. The Python code interfaces with PSCAD/EMTDC to process real-time measurements, which are then input into the models to detect potential cyber intrusions within each time window.

1) *Detection of Attack on Load Buses:* In this case study, the targeted buses are load buses, and both the real and reactive power of those buses are manipulated. The magnitude of manipulation applied to the targeted buses is randomly selected from the interval $[-0.15, -0.1] \cup [0.1, 0.15]$. Table II presents a comparison of the performance of the proposed model against baseline models in this attack scenario. The proposed model consistently outperforms the baseline models across all evaluation metrics. This indicates a higher capability of the proposed model to detect AC FDI attacks accurately while effectively minimizing false positives and false negatives. The CNN-LSTM with UKF model shows moderate improvement over CNN-LSTM, CNN, and SVM but falls short of the graph-based models (GACN and GACN-LSTM), highlighting the importance of incorporating physics-informed spatial correlations in the detection process. Ultimately, the superior performance of the proposed model highlights the capabilities of this combined approach in detecting and localizing AC FDI attacks.

2) *Attack Detection of Generator Buses:* In this case study, the targeted buses are generator buses, and the real power and voltage of these buses are manipulated, with the intensity of manipulation being identical to that in the first case study. Table III presents the performance comparison of the proposed approach and baseline models in the second attack scenario. Consistent with the results from the first scenario, the proposed model outperforms all baseline models across all evaluation metrics. While slight reductions are observed in some metrics compared to the first attack scenario, the proposed method maintains a high level of performance, demonstrating its robustness even under varied attack patterns. The slight decrease in performance across all models suggests that this attack scenario presents a greater challenge for detection. Among the baseline models, the GACN-LSTM and GACN models perform relatively well, with GACN-LSTM showing slightly better results than GACN in most metrics. However, both

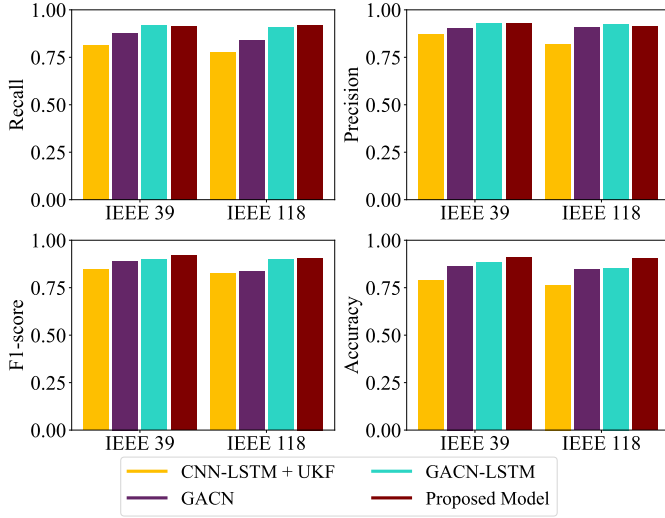


Fig. 2. Comparison of CNN-LSTM + UKF [11], GACN, GACN-LSTM and the proposed method across various metrics including recall, precision, F1-score, and accuracy.

models still lag behind the proposed method, particularly in precision and F1-score, highlighting the effectiveness of the integrated approach. On the other hand, simpler models like SVM and CNN exhibit significantly lower performance, with notable reductions in recall and F1-score, emphasizing their limitations in accurately identifying AC FDI attacks in complex scenarios.

3) *Combined Attack Detection*: The targets of this case study are both load and generator buses, and either the real and reactive power for load buses or the real power and voltage for generator buses are manipulated, with the intensity of manipulation consistent with previous case studies. Table V and Fig. 2 outline the performance comparison of the proposed model and the baseline models in the third attack scenario. As observed in previous scenarios, the model exhibits superior precision and recall, further confirming its ability to accurately detect AC FDI attacks. Compared to the first and second attack scenarios, the proposed method exhibits slightly weaker performance in the third scenario, particularly in precision and F1-score. However, the proposed model performs better than baseline models on every metric. This suggests that the model is highly robust, even in scenarios that combine both load and generator bus attacks, which typically present more complex and dynamic challenges. Despite varying attack patterns, the model precision remains high, indicating it can maintain low false positive rates.

4) *Combined Attack Detection with Higher Intensity Attack*: To further evaluate the contributions of the proposed framework, another case study was conducted using the third case study scenario with a modification. In this study, the magnitude of the manipulation applied to the targeted buses was randomly selected from a wider range, specifically from the interval $[-0.2, -0.05] \cup [0.05, 0.2]$. This modification introduced more variability in the severity of the attacks, testing the robustness of the detection models under more diverse conditions. As shown in Table VI and Fig. 3, even in the presence of this

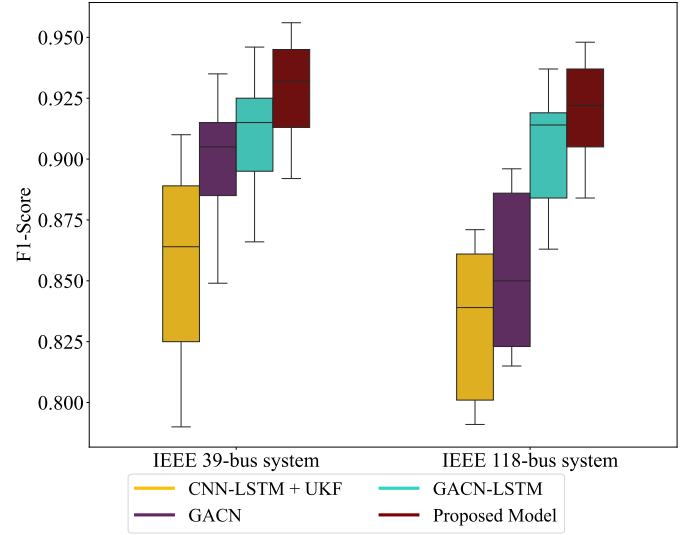


Fig. 3. Comparison of CNN-LSTM + UKF [11], GACN, GACN-LSTM and proposed method F1-score for the two test cases.

broader range of cyber intrusions, the proposed method has a better performance compared with the baseline models across all metrics. This is particularly evident in the precision and F1-score, highlighting the model's ability to consistently detect attacks, even with varying conditions and attack patterns.

The results of the performance evaluation clearly indicate that the proposed approach offers significant improvements over base methods, particularly in terms of precision, recall, and interpretability. A key advantage of this approach lies in its use of the attention mechanism, which prioritizes the most relevant connections or nodes based on the current state of the system and the patterns learned from historical data. This significantly enhances interpretability, as the model does not treat the power grid as a black box but instead provides insights into the real-world interactions and dependencies within the grid. By leveraging the physical topology of the power grid, the graph attention mechanism helps localize attacks with high accuracy, enabling operators to visualize which parts of the grid are most at risk. This not only facilitates early detection of attacks but also supports preemptive actions to mitigate potential disruptions. In contrast to previous deep learning approaches, which often lack transparency, the proposed method offers interpretable results that align with the physical structure of the grid, providing a more actionable and intuitive framework for cyberattack detection and localization.

The integration of graph-based learning, temporal sequence analysis, and state estimation through the UKF enhances the model's robustness in detecting and mitigating AC FDI attacks. The model's ability to achieve high F1-score demonstrates its effectiveness in distinguishing between normal and compromised states within power grid environments. However, despite these promising results, some limitations were observed, particularly regarding the increased computational time required for training due to the model's complexity. Future work could focus on optimizing the training process to reduce

TABLE V
PERFORMANCE COMPARISON OF THE PROPOSED MODEL AND BASELINE MODELS IN THE THIRD ATTACK SCENARIO

Model	IEEE 39-Bus System				IEEE 118-Bus System			
	Accuracy	Precision	Recall	F1-Score	Accuracy	Precision	Recall	F1-Score
SVM	0.611	0.591	0.497	0.527	0.590	0.469	0.355	0.402
CNN	0.697	0.715	0.577	0.603	0.641	0.620	0.491	0.526
CNN-LSTM [11]	0.721	0.729	0.671	0.667	0.689	0.726	0.676	0.612
CNN-LSTM + UKF [11]	0.786	0.870	0.813	0.848	0.764	0.817	0.777	0.823
GACN	0.864	0.904	0.879	0.891	0.848	0.906	0.841	0.836
GACN-LSTM	0.881	0.928	0.917	0.899	0.854	0.925	0.908	0.900
Proposed Model	0.910	0.932	0.913	0.918	0.903	0.916	0.917	0.906

TABLE VI
PERFORMANCE COMPARISON OF THE PROPOSED MODEL AND BASELINE MODELS IN THE ABLATION STUDY

Model	IEEE 39-Bus System				IEEE 118-Bus System			
	Accuracy	Precision	Recall	F1-Score	Accuracy	Precision	Recall	F1-Score
SVM	0.627	0.605	0.503	0.540	0.604	0.486	0.368	0.419
CNN	0.713	0.729	0.593	0.619	0.654	0.636	0.507	0.539
CNN-LSTM [11]	0.738	0.743	0.685	0.681	0.705	0.740	0.693	0.629
CNN-LSTM + UKF [11]	0.800	0.886	0.827	0.864	0.778	0.833	0.793	0.839
GACN	0.880	0.920	0.895	0.905	0.862	0.922	0.857	0.850
GACN-LSTM	0.895	0.944	0.931	0.915	0.870	0.941	0.924	0.914
Proposed Model	0.924	0.946	0.929	0.932	0.917	0.932	0.931	0.922

computational overhead and improve the model's scalability for larger the power grid.

F. Attention Interpretability

The attention mechanism is not only used internally for feature weighting, but also provides an interpretable signal for operators. In practice, the normalized attention coefficients can be aggregated over a detection window and ranked to identify the most affected buses. This yields a simple "top- k suspect list," which can be overlaid on the grid topology to highlight the regions most likely under attack. As illustrated in Fig. 4, the attention weights concentrate on the attacked buses and their immediate neighbors.

G. Practical Considerations for Real-World Deployment

In this paper, the dataset is derived from PSCAD/EMTDC simulations. In real-world applications, PMU measurements may include higher noise levels, missing samples, or unexpected disturbances not captured in the simulation. Our framework partially addresses these challenges through the UKF, which down-weights inconsistent measurements via the innovation covariance and thus provides robustness against measurement noise. Missing data can also be mitigated with standard interpolation or imputation methods before being input to the GACN-LSTM. A thorough validation with field PMU data remains an important next step, and forms a direction for future work to ensure reliable deployment.

Another key practical challenge is grid adaptability. In practice, grid topologies change due to switching operations, reconfiguration, and variations in generation dispatch. The proposed framework mitigates this challenge by embedding the physical topology into the GACN layer, so re-training on a modified grid does not require fundamental changes to the

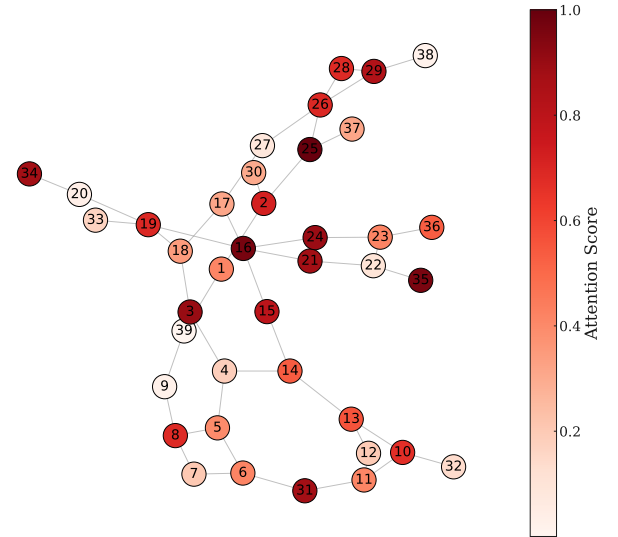


Fig. 4. Visualization of attention weights during a representative AC-FDI attack on the IEEE 39-bus system.

architecture. Because GACN takes the grid adjacency matrix as an explicit input, topology changes can be incorporated by updating the edge admittances and re-running inference. For small or localized changes, GACN parameters remain useful, and only minimal fine-tuning of the final layers is typically required. Similarly, for the UKF layer, topology or dispatch updates imply adjustments to the measurement function and measurement noise covariances, which can be updated without retraining.

To further enhance adaptability, several practical online and transfer learning techniques can be considered in future work. These include fine-tuning the classifier on a small buffer

of recent measurements after a topology change, continual learning with replay to avoid catastrophic forgetting, and periodic retraining during low-load windows using logged measurements combined with simulated events for the new topology. These strategies will help ensure that the framework remains robust and effective under dynamic and evolving grid conditions.

V. CONCLUSION

This paper proposed a novel approach for detecting AC FDI attacks in the power grid, combining GACN, LSTM networks, and UKF. By integrating graph-based modeling, temporal sequence analysis, and dynamic state estimation, the framework enhances accuracy, scalability, and interpretability in detecting and localizing AC FDI attacks. Simulation results on the IEEE 39-bus and 118-bus systems demonstrate the method's robustness. GACN effectively captures spatial correlations, LSTM models temporal patterns, and UKF increases resilience by accurately estimating the grid's true state under noisy conditions. The attention mechanism in GACN further improves interpretability, highlighting critical nodes and connections vulnerable to attacks. This work emphasizes the importance of combining deep learning with traditional state estimation for securing the power grid. Future research could focus on optimizing computational efficiency and extending the framework to larger grids and other types of cyberattacks.

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